René García Torres

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Education

EIT InnoEnergy Double Degree Master
M.Sc. Smart Electrical Networks and Systems (SENSE) *2025 – 2027*

- **KTH Royal Institute of Technology** — Master of Science *Stockholm, Sweden | Aug. 2025 – Jul. 2026*
- **ESADE Business School** — Innovation & Entrepreneurship Course *Barcelona, Spain | Feb. 2026 – Jul. 2026*
- **UPC Universitat Politècnica de Catalunya** — Master's in Energy Engineering *Barcelona, Spain | Sep. 2026 – Jul. 2027*

Tecnológico de Monterrey Monterrey, Mexico.
B.S. Electronics Engineering, Cyber-Physical Systems Minor *Aug. 2019 – Jul. 2023*

***GPA**: 96/100. Awarded by the National CENEVAL Organization due to high score on exit exam.*

Core Competencies

Electrical Machines & Drives: dq modelling and parameter identification of DC, induction and PMSM machines; field-oriented control; IPM traction motor sizing with FEM validation (COMSOL).
Power Electronics & Electromobility: Design and control of power converters and motor drives; loss and thermal design; Double Pulse Testing; 2-/3-level EV traction inverters (IGBT vs. SiC); PFC and STATCOM.
Embedded Systems: C firmware for ESP32 (FreeRTOS), dsPIC33 and automotive ECUs (AUTOSAR, CAN/LIN).
Power Systems & Grid Integration: Load-flow and short-circuit analysis, micro-grids, PV and wind integration, power quality, HVDC.
Programming: C, Rust and ASM for embedded systems; Python, MATLAB and Java for simulation tools and automation.
Innovation & Business: New venture creation, business model design, customer validation, financial planning and investor pitching.

Languages

Spanish: Native, **English**: C1 (Proficient), **German**: A1 (Basic)

Experience

Aptiv Monterrey, México
Embedded Software Engineer *Jul. 2023 – Jul. 2025*

- Developed AUTOSAR automotive ECU software following the V-model, from requirements documentation to qualification testing.
- Executed SIL and HIL testing and debugging with Lauterbach and Vector Informatik tools.
- Developed CAN/LIN gateways, DTCs and DIDs, and integrated supplier software components.

Tecnológico de Monterrey Monterrey, México
Undergraduate Research Assistant *Jun. 2022 – Jul. 2023*

- Developed a three-phase motor control system using a dsPIC33 microcontroller, with programming in Simulink and C.
- Designed and constructed buck, boost, and buck-boost converter PCBs using Altium Designer.
- Led a project on tiny, inexpensive solar cars aimed at engaging future university students in the electronics field.

Technological Investigation and Innovation Park (PIIT) Apodaca, México
Research Summer Intern *Jun. 2022 – Jul. 2022*

- Developed an ESP32/FreeRTOS smart microbiological incubator with computer-controlled light, temperature and sound.

Projects

IPMSM Motor Design for Formula Student | *Motor Design, FEA, MATLAB, COMSOL* Mar. 2026 – Aug. 2026
Sized a V-shape IPM rotor in a unit-tested MATLAB library, auto-generating COMSOL electromagnetic and structural studies.

Electro-thermal Analysis of Power Switches (Volvo Cars) | *Python, SPICE, PLECS* Mar. 2026 – Jul. 2026
Built a Python Double Pulse Test pipeline (6–8× faster) to compare IGBT/SiC losses in 2-/3-level EV inverters.

stackCOM: Modular STATCOM Venture (ESADE) | *Business Development, FACTS, Grid Codes* Feb. 2026 – Jul. 2026
CEO of a five-person team building stackable 0.5 MVAr STATCOM modules for 33 kV renewable plant connections.

Keyrings Assembly Line Automation | *IIoT, ROS, Robot Vision, Neural Networks, Cobots* Aug. 2022 – Dec. 2022
Developed a fully functional automated short assembly line for keyrings. Used Web APIs, computer vision and mobile robots.

Courses & Certifications

Accelerated Computer Science Fundamentals (UIUC) | C++ Essential Training (LinkedIn) | C++ Templates and the STL (LinkedIn) | Introduction to Power Electronics (CU Boulder) | Introductory C Programming (Duke) | Java Programming: Solving Problems with Software (Duke) | Parallel Programming in Java (Rice) | Python for Everybody (U. Michigan) | Python Object-Oriented Programming (LinkedIn) | Rust Fundamentals (Coursera) | Training Neural Networks in C++ (LinkedIn)

Technical Skills

Hardware Skills: Soldering (PTH and SMT), oscilloscope, multimeter, waveform generator, drill, logic analyzer.
Programming Languages: Bash, C/C++, FreeRTOS, Java, MATLAB, Makefile, Python, R, Rust, Simulink.
Simulation & Design Tools: COMSOL Multiphysics, PLECS, PSpice, LTspice, ngspice, KiCad, Altium Designer, MATLAB/Simulink.
Software Tools: ESP-IDF, CMake, Lauterbach, Vector tools, Docker, Git, Gerrit, Jira, LaTeX, ROS, GNU/Linux.